

Differential Equation

3. Homogeneous and Reducible: Reduction to variable separable form

By transformations certain non-separable ODEs can be made separable. It is done by introducing y a new unknown function. Let,

$$y' = f\left(\frac{y}{x}\right)$$

Put $y = ux$ and taking the differentiation $y' = u'x + u$

Substitution into $y' = f\left(\frac{y}{x}\right)$ then gives $\frac{du}{f(u)-u} = \frac{dx}{x}$

Example: Solve the ODE $2xyy' = y^2 - x^2$.

Solution: We divide the given differential equation by $2xy$, obtaining

$$y' = \frac{y^2}{2xy} - \frac{x^2}{2xy} = \frac{1}{2} \left(\frac{y}{x} - \frac{x}{y} \right).$$

Now $y' = u'x + u$, so that we obtain

$$u'x + u = \frac{1}{2} \left(u - \frac{1}{u} \right),$$

hence by simplification,

$$u'x = -\frac{1}{2} \left(u + \frac{1}{u} \right) = -\frac{u^2 + 1}{2u}.$$

We see that separation of variables now gives

$$\frac{2udu}{1+u^2} = -\frac{dx}{x}.$$

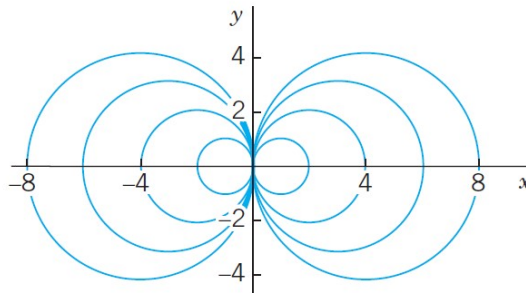
By integration,

$$\ln(1+u^2) = -\ln|x| + c^* = \ln \frac{1}{|x|} + c^*, \quad 1+u^2 = \frac{c}{x},$$

Replacing u by y/x gives $1 + (y/x)^2 = c/x$. We multiply by x^2 to get

$$x^2 + y^2 = cx. \quad \left(x - \frac{c}{2} \right)^2 + y^2 = \frac{c^2}{4},$$

This general solution represents the family of circles with centers on the x-axis and all passing through the origin.



Example: Transformations $v = ay + bx + k$

Sometimes a transformation $v = ay + bx + k$ may lead to a separable differential equation. We illustrate the idea of this method by a simple equation,

$$(2x - 4y + 5)y' + x - 2y + 3 = 0.$$

Solution: The terms $2x - 4y$ and $x - 2y$ suggest that we try $v = x - 2y$. Then $2y = x - v$ and $y' = (1 - v')/2$.

By substitution and simplification, the given equation become

$$(2v+5)v' = 4v+11.$$

Multiplying by 2 and separating variables, we obtain

$$\frac{4v+10}{4v+11} dv = \frac{4v+11-1}{4v+11} dv = \left(1 - \frac{1}{4v+11}\right) dv = 2 dx.$$

By integration,

$$v - \frac{1}{4} \ln|4v+11| = 2x + c^*.$$

Since $v = x - 2y$, this implicit general solution of the given equation may be written

$$4x + 8y + \ln|4x - 8y + 11| = c$$